

Lab Problem Statement 6: Customer Segmentation and Predictive Analytics Using Machine Learning

Problem Statement:

An e-commerce company wants to better understand its customers and improve its marketing decisions. Using historical online retail transaction data, develop a machine-learning-based analytics solution that identifies meaningful customer segments and predicts whether a customer is likely to belong to a high-value customer category.

Students shall preprocess the transactional data and construct suitable customer-level features such as **Recency, Frequency, Monetary Value (RFM), average transaction value, quantity purchased, and purchase frequency**. Apply appropriate techniques for handling missing values, cancellations, outliers, and feature scaling.

Perform **customer segmentation using K-Means clustering**, determine the optimal number of clusters using the **Elbow Method**, and compare the obtained segmentation with **Hierarchical Clustering** using a dendrogram. Evaluate the quality of clustering using the **Silhouette Score**.

Apply **Principal Component Analysis (PCA)** for dimensionality reduction and visualize the identified customer segments using a **2D PCA projection**. Develop appropriate customer profiles for each cluster based on their purchasing characteristics.

Based on the generated clusters, identify an appropriate **high-value customer segment** and build at least **two supervised machine learning models** (such as Random Forest and SVM) to predict whether a customer belongs to this segment. Evaluate and compare the models using suitable classification metrics such as **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix**.

Finally, interpret the clustering and prediction results and recommend suitable **marketing strategies for different customer segments**.

Dataset: UCI Machine Learning Repository – Online Retail Dataset

[UCI Online Retail Dataset](#)

The dataset contains **541,909 online retail transactions** and includes attributes such as Invoice Number, Product, Quantity, Invoice Date, Unit Price, Customer ID, and Country. It is explicitly associated with clustering tasks, making it suitable for this experiment.

Expected Outputs / Deliverables:

1. Data preprocessing and customer-level feature engineering.
2. Elbow Method graph for selecting the optimal number of clusters.
3. K-Means customer segmentation.
4. Hierarchical clustering and dendrogram.
5. Silhouette Score and comparison of clustering results.
6. PCA-based 2D visualization of customer segments.
7. Cluster-wise customer profiling and interpretation.
8. Random Forest and SVM models for high-value customer prediction.

9. Performance comparison using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.
10. Feature importance analysis for the predictive model.
11. **Interactive 3D visualization of customer clusters using Plotly.**
12. Segment-wise targeted marketing recommendations.

Submission Instructions

- Complete the entire experiment using **Python in Google Colab**.
- Submit the **Colab Notebook in PDF format**.
- Upload the complete notebook/code to your **Git repository**.
- Include the **Git repository link** in the final submission.
- The notebook should clearly contain the problem statement, dataset description, preprocessing, implementation, visualizations, results, interpretation, and conclusion.

AI Feedback Scores

Rating: 1 – Lowest | 4 – Highest

Parameter	Score
Real-life Application	4 / 4
Problem-solving Ability	4 / 4
Critical Reasoning	4 / 4
Creativity	4 / 4
Alignment with Learning / Course Outcomes	4 / 4

Estimated Time Required

1 Hour 45 Minutes – 2 Hours

Lab 3: 50–55 Minutes

- Preprocess customer data through standardization and outlier treatment
- Perform K-Means and Hierarchical Clustering
- Validate clustering results using Silhouette Scores and PCA
- Visualize clusters and prepare customer segmentation recommendations

Lab 4: 55–65 Minutes

- Perform feature engineering using sensor data
- Apply PCA and t-SNE for dimensionality reduction and visualization

- Build predictive maintenance models using Random Forest and SVM
- Evaluate model performance and prepare deployment-oriented maintenance recommendations

COs Linked with the Assignment

CO3, CO4, CO5, CO6

LOs Linked with the Assignment

LO 4.2, LO 5.1, LO 5.2, LO 6.1, LO 6.2, LO 6.3

Justification

This laboratory assignment provides students with advanced practical exposure to **unsupervised learning, predictive analytics, dimensionality reduction, and industrial machine learning applications** using realistic customer analytics and predictive maintenance scenarios. The problem statements closely resemble modern data science projects where organizations leverage clustering techniques, feature engineering, machine learning models, and visualization methods to support strategic decision-making and operational optimization.

The first laboratory introduces students to **unsupervised machine learning** by requiring them to perform customer segmentation using **K-Means** and **Hierarchical Clustering** techniques. Students preprocess customer data through standardization and outlier treatment, determine the optimal number of clusters using the elbow method, validate clustering quality through **Silhouette Scores**, perform cluster profiling using **Principal Component Analysis (PCA)**, and create interactive visualizations. These activities strengthen students' understanding of exploratory data analysis, cluster validation, and business-driven customer segmentation strategies.

The second laboratory extends these concepts to an **industrial predictive maintenance** scenario where students engineer features from sensor data, perform dimensionality reduction using **PCA** and **t-SNE**, and develop predictive models using **Random Forest** and **Support Vector Machines (SVM)**. Students further evaluate model effectiveness using precision-oriented performance measures and early warning analysis while designing deployment-oriented maintenance recommendations. These activities simulate industrial predictive maintenance systems widely adopted in Industry 4.0 environments.

The assignment demonstrates strong alignment with **CO3** by requiring students to implement advanced machine learning techniques and predictive analytics using R. It directly addresses **CO4** through comprehensive model evaluation, feature engineering, dimensionality reduction, and performance optimization. The comparison and application of multiple machine learning algorithms across supervised and unsupervised learning tasks significantly contribute to **CO5**, enabling students to select appropriate analytical techniques for diverse real-world datasets. Furthermore, the complete workflow encompassing data preprocessing, feature engineering, clustering, predictive modeling, visualization, evaluation, and deployment planning strongly supports **CO6**, allowing students to design and implement end-to-end data science solutions using industry-standard practices. The inclusion of interactive visualizations, predictive maintenance analytics, customer segmentation strategies, and deployment blueprints makes

this assignment highly relevant to contemporary artificial intelligence, manufacturing analytics, and business intelligence applications.

AI Tool Used

ChatGPT

Prompt(s) Used

"Evaluate the given R Programming laboratory assignment using a 1–4 rubric for Real-life Application, Problem-solving Ability, Critical Reasoning, Creativity, and Alignment with Learning / Course Outcomes. Identify the estimated time required, map the relevant Course Outcomes (CO1–CO6) and Learning Outcomes (LOs), and justify the ratings based on the R Programming syllabus covering unsupervised learning, clustering, dimensionality reduction, feature engineering, predictive maintenance, supervised machine learning, model evaluation, visualization, and end-to-end data science workflows."